CLAIMAUDIT

Invention disclosure
Research response draft

Mapped to VIT IDF-B | 19 October 2026

19 October 2026. Response annex mapped to all ten questions in VIT IDF-B, Document 02-IPR-R003, Issue 2 / 01.02.2024. Research prototype implemented; patent novelty and inventive step remain unresolved. This annex preserves the original blank form. The 18 October draft describes an earlier proposal and is superseded for implementation status.

1. Title of the invention

Context-Consistent Scientific Claim Verification with Correlated Source Interpretations and Decision-Sufficient Clarification.

Working implementation name: ScopeGuard, an experimental extension to ClaimAudit. The title does not assert patentability.

2. Field / area of invention

Scientific document analysis, evidence-grounded natural-language systems, structured numerical verification and research-assistant software. The immediate use case is checking whether a generated scientific comparison combines measurements from incompatible experiments or alternative interpretations of table metadata.

3. Prior patents and publications from literature

Prior art	Relevant established mechanism	Distinction to investigate
RefChecker, 2024 [1]	Fine-grained claims and evidence verification	Preserve shared experimental context across predicates
Proof-Carrying Numbers, 2025 [2]	Source-referenced numerical verification	Joint context compatibility, not numerical matching alone
When Verification Fails, 2026 [3]	Failure on compound scientific constraints	Correlated source alternatives and exact binding analysis
EG-VAR, 2026 [4]	Tool-grounded proofs and evidence scope	Bounded alternative bundles and header clarification
Certain answers, 2016 [5]	Certainty over incomplete information	Application-specific implementation; foundations already known
AmbiFC, 2021 [6]	Ambiguous claims and evidence-based verdicts	Source metadata alternatives, not ambiguity alone
Claim Cards, 14 Oct 2026 [7]	Explicit scientific interpretations and verdict changes	All enumerated source bindings and decision-sufficient acquisition
US20260127386A1 [8]	Language-model constraint/theory solving	Narrow source-binding mechanism requires claim-level review
US20250077940A1 [9]	Verification and conditional answer output	Generic verification and retry are not claimed new

These differences are hypotheses, not findings of novelty or non-infringement. Exact joins, dynamic programming, provenance and uncertainty reasoning are established. Search is preliminary and not a patent clearance opinion.

4. Summary and background of the invention

ClaimAudit already compares embeddings, NLI, and NLI combined with numerical/comparison checks. The remaining problem is evidence splicing: independently valid measurements can be combined into a claim that no single experiment supports. Ambiguous source metadata can also produce mutually exclusive interpretations that must not be pooled.

Example: configuration X reports 90% accuracy at 22 ms, while Y reports 80% accuracy at 2 ms. The claim that one configuration achieves at least 90% accuracy at at most 2 ms has no joint witness. Complete declared experimental coverage allows contradiction; partial evidence only permits an insufficient-evidence result.

The candidate contribution is a bounded verification workflow that retains correlated source alternatives, checks joint support across all admissible interpretations, merges decision-equivalent states, and identifies a source clarification sufficient to remove verdict ambiguity. Its distinctiveness over the cited work is unproven. The implemented result is an engineering proof of concept, not a new general reasoning algorithm.

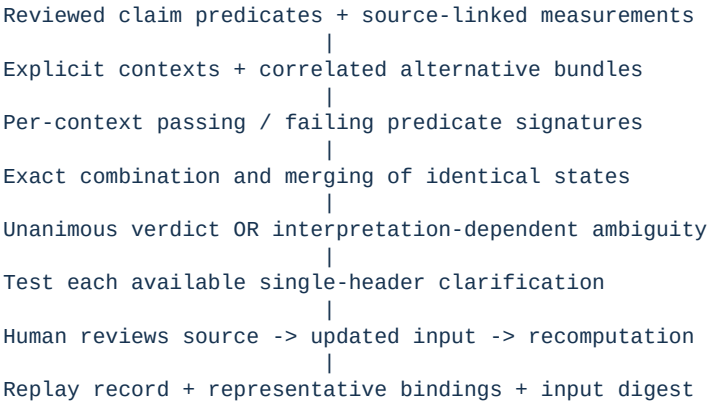
5. Objectives of invention

- Detect cross-experiment and mutually exclusive evidence splicing.
- Avoid converting missing measurements into contradictions.
- Preserve uncertainty when plausible source interpretations produce different decisions.
- Reduce repeated computation for interpretations with identical satisfaction states.
- Identify a single obtainable metadata clarification that settles interpretation-dependent decisions.
- Produce reproducible records and a transparent researcher-facing demonstration.

6. Working principle in brief

A reviewer supplies numerical claim predicates, context IDs, source-linked facts and mutually exclusive alternative bundles. The system checks the whole conjunction in each interpretation through an exact state-merging algorithm. It returns supported, contradicted, insufficient or ambiguous, with incomplete on a state-budget limit. For ambiguity, it conditionally resolves each header and recommends the cheapest header whose every outcome settles the decision. A person reviews the source; the engine recomputes the result.

7. Description of the invention in detail



Input values use exact decimal strings and exact units. Each claim asserts that one listed context meets all threshold predicates. Exactly one alternative in each block applies; correlated uncertainties must reside in the same block. The current implementation does not extract these records from arbitrary PDFs or support universal claims, cross-context comparisons, statistical significance or implicit unit conversion.

A state holds passing and failing predicate masks for each context. Combining facts is bitwise union; merging identical masks preserves all future decisions. The engine counts merged interpretations and retains representative bindings. Counts are not probabilities. Worst-case state growth remains exponential, and exceeding the configured cap produces no substantive verdict.

Contradiction requires an explicit exhaustive-domain declaration and an observed failing condition in every context. Potentially coexisting conflicting measurements are rejected as invalid input. The replay digest identifies the structured input; it does not prove source authenticity or constitute a formal proof certificate.

Clarification costs are declared positive integers. The implemented acquisition policy selects a cheapest sufficient single clarification; it is not an optimal multi-step policy. Retrieval, automatic header interpretation and incremental dependency invalidation are not implemented.

8. Experimental validation results

Measured locally on 19 October 2026. The new dataset has 300 synthetic cases from six parameterized families. It is a controlled engineering check, with no independent annotation or held-out paper evaluation.

Method	Support accuracy	Qualification
Atomic matching	50.0%	Discards context
Flattened context join	66.7%	Discards alternative exclusivity
Exhaustive verifier	100.0%	Strong exact reference
ScopeGuard	100.0%	Same correctness as exact reference

All 300 intended four-way statuses and exhaustive outcome counts agree. A further 200 seeded random problems are checked against an independently implemented exhaustive reference in tests. A favorable stress example represents 1,048,576 interpretations with one signature per layer and 40 transitions; this is not a general speedup measurement.

Tests cover decimal equality, units, missing evidence, conflicting records, resource-limit abstention, replay modification detection, representative bindings, clarification selection and all six interactive scenarios. The validation manifest and reproducible commands are saved with the project. No new embedding/NLI accuracy comparison is claimed from this structured-input experiment.

Real-paper annotation, extraction-quality measurement, paper-level test splits, strong language-model comparison, adverse scaling cases and acquisition-cost evaluation remain necessary for a journal submission.

9. What aspects of the invention need protection?

Candidate for professional assessment: the specific sequence of correlated source-binding representation, exact joint-support evaluation, verdict-equivalent state merging, and decision-sufficient metadata clarification for scientific document verification. Any proposed claim must identify a technical distinction over existing uncertain-database reasoning, Claim Cards, numerical verification and solver-assisted systems.

Do not claim ownership of embeddings, NLI, generic numerical checks, all-world reasoning, dynamic programming, evidence provenance, clarification generally or cryptographic hashes. No claim set is proposed as filing-ready. Novelty, inventive step, industrial applicability and software eligibility require separate assessment; a completed disclosure is not patent approval.

The implemented prototype and its reproducible artifacts can support an institutional IP-cell discussion. Patentability remains unresolved rather than positively established by this search.

10. Technology readiness level

Suggested self-assessment: TRL 3, experimental proof of concept. A bounded engine, synthetic verification suite and interactive local demo exist. It has not been validated on independently annotated real scientific papers or in an operational research workflow. TRL 4 or higher is not justified by the current evidence.

References

[1] RefChecker: https://arxiv.org/abs/2405.14486

[2] Proof-Carrying Numbers: https://arxiv.org/abs/2509.06902

[3] When Verification Fails: https://arxiv.org/abs/2604.10990

[4] EG-VAR: https://arxiv.org/abs/2607.12650

[5] Certain answers as objects and knowledge: https://homepages.inf.ed.ac.uk/libkin/papers/aij.pdf

[6] AmbiFC: https://arxiv.org/abs/2104.00640

[7] Same Claim, Different Interpretations: https://arxiv.org/abs/2610.20457

[8] Language model theory solvers: https://patents.google.com/patent/US20260127386A1/en

[9] Verification and conditional output: https://patents.google.com/patent/US20250077940A1/en

[10] Indian Patent Office, patent basics: https://ipindia.gov.in/basics-of-patents